COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS

GRAPH-BASED QUERYING AND MUTATIONS OF LINKED DATA

Master thesis

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Bc. Ondrej Maslen

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Master thesis

Study program:	Applied informatics
Branch of study:	Applied informatics
Department:	Department of Applied Informatics
Supervisor:	Mgr. Ján Kľuka, PhD..
Consultant:	

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Bc. Ondrej Maslen

49092252

Univerzita Komenského v Bratislave
Fakulta matematiky, fyziky a informatiky

ZADANIE ZÁVEREČNEJ PRÁCE

Meno a priezvisko študenta:	Bc. Ondrej Maslen
Študijný program:	aplikovaná informatika (Jednoodborové štúdium, magisterský II. st., denná forma)
Študijný odbor:	informatika
Typ záverečnej práce:	diplomová
Jazyk záverečnej práce:	anglický
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Názov: Graph-based querying and mutations of Linked Data
Grafové dopyty a modifikácia prepojených dát

Anotácia: Súčasným trendom vo vývoji webových aplikácií je prístup k dátam pomocou grafových dopytov komponovaných na strane klienta v jazykoch ako GraphQL a JSON-LD-Query. V predchádzajúcej práci nášho tímu [1] sme vyvinuli podporný výučbový systém (LMS), ktorého dátová vrstva využíva technológie prepojených dát (Linked Data, LD) – RDF, OWL2 a SPARQL [3], no prezentačná vrstva k nim pristupuje cez rozhranie (API) typu REST implementované špecializovaným midlvérom [2], ktorého úlohou je aj autentifikácia a autorizácia. Toto API by sme chceli nahradiť expresívnejším API na báze GraphQL.

Cieľ:
• Analyzovať existujúce REST-ové API LMS a jeho dátový model [1, 2]
• Naštudovať predchádzajúce výsledky v dopytovaní LD pomocou GraphQL [4, 5]
• Vyhodnotiť ich aplikovateľnosť na prípad LMS
• Aplikovať, modifikovať, a rozšíriť dostupnú implementáciu rozhrania medzi LD a GraphQL podľa potrieb nášho prípadu použitia

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Vedúci:	Mgr. Ján Kľuka, PhD.
Katedra:	FMFI.KAI - Katedra aplikovanej informatiky
Vedúci katedry:	doc. RNDr. Tatiana Jajcayová, PhD.
Dátum zadania:	21.10.2018

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49092252

Univerzita Komenského v Bratislave
Fakulta matematiky, fyziky a informatiky

garant študijného programu

študent

vedúci práce

I hereby declare that I have written this thesis by myself, only with help of referenced literature, under the careful supervision of my thesis advisor.

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Acknowledgement

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Abstract

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Kľúčové slová:

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Terminology

Terms

- **Star field tracking (sidereal)**
  Ground-based tracking mode in which, telescope is moving in the same direction and speed as the apparent motion of stars.
- **Object tracking**
  Tracking mode, where the focus is aimed at the moving object of interest and the telescope is moving in the same way.
- **Survey**
  Observation of a region of the sky when no specific target is defined.
- **Star catalog**
  A list of stars with its positions and magnitude.
- **Star tracker**
  An optical device usually used to determine the orientation of satellite using positions of the stars.
- **Deblending**
  The process of separating overlapping objects.

Abbreviations

- **CCD** - Charge-Coupled Device.
- **IAA** - International Academy of Astronautics.
- **USSSN** - US Space Surveillance Network.
- **CNN** - Convolutional Neural Network.
- **FC** - Fully-Connected.
- **RSO** - Resident Space Object.
- **ML** - Machine Learning.

- **SDSS** - Sloan Digital Sky Survey.
- **PCA** - Principal Component Analysis.
- **ANN** - Artificial Neural Network.
- **NN** - Neural Network.
- **MLP** - Multi-Layer Perceptron.
- **R-CNN** - Region-based Neural Network.
- **MS COCO** - Microsoft Common Objects in Context.
- **AGO** - Astronomical and Geophysical Observatory in Modra.
- **AGO70** - The Newtonian telescope at AGO, with 70 cm parabolic mirror.
- **ESA** - European Space Agency.
- **PECS** - Plan for the European Cooperating States.
- **FMPI** - Faculty of Mathematics, Physics and Informatics.
- **FITS** - Flexible Image Transform System.
- **RADEC** - Right Ascension and Declination.
- **FOV** - Field Of View.
- **PSF** - Point-Spread Function.
- **FWHM** - Full Width at Half Maximum.
- **ADU** - Analogue-to-Digital Unit.
- **ADC** - Analog to Digital Converter.
- **SVM** - Support-Vector Machine.
- **ResNet** - Residual Neural Network.
- **ILSVRC** - ImageNet Large Scale Visual Recognition Challenge.
- **RELU** - Rectified Linear Unit.
- **TSV** - Tab-Separated Values.
- **CLI** - Command Line Interface.
- **YAML** - YAML Ain't Markup Language.

Motivation

Motivation

Conclusion

Conclusion

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